

ASHAR AYUB

Lead Software Engineer · Distributed Systems, Real-Time Platforms, Production LLM Integration

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SUMMARY

Lead software engineer with 5+ years building and operating backend infrastructure and real-time systems. Recent work includes an emergency multi-region AWS migration after physical data centre loss, a custom Coturn/WebSocket stack serving a 250+ device fleet, and production LLM integration in clinical training and telemedicine platforms. Lead architectural review of both human-authored and AI-generated code, and mentor junior engineers. Comfortable owning a system end to end — architecture, infrastructure, and production incidents — rather than a single layer of it.

CORE SKILLS

Architecture & Cloud AWS (EC2, RDS, S3, Load Balancers, multi-region), Azure, GCP, Cloudflare, MinIO, Nginx, Ubuntu/VPS, Laravel Forge, CI/CD (GitHub Actions), zero-downtime deployment.

Backend PHP/Laravel, Node.js, NestJS, Python/FastAPI, REST API, GraphQL.

Real-Time & Data WebSockets, Coturn (TURN/STUN), Redis, MySQL, PostgreSQL, MongoDB, query tuning & indexing, caching, queues, Crons.

AI in Production LLM integration (FastAPI, vector DBs), agentic workflows, Model Context Protocol (MCP), AI-generated code review & architectural governance.

Frontend & Mobile Vue/Inertia.js, React, Angular, Ionic/Capacitor, React Native, Tailwind CSS, Kotlin, Java.

PROFESSIONAL EXPERIENCE

Lead Developer — KLABS Tech Solutions Jun 2021 – Present · *Karachi, PK*

- **Architectural governance:** Lead server-side architecture across concurrent client engagements; conduct design and code review of both human-authored and AI-generated components, owning correctness, security, and production accountability.
- **Mentorship:** Mentored 5+ junior engineers across backend, mobile, and development work; established review standards and delivery practices for the team.
- **Platform engineering:** Architect RESTful services in PHP/Laravel and Node.js. Eliminated N+1 query patterns and added composite indexing on the fleet telemetry endpoint, cutting down API response time from seconds to ms.
- **Real-time systems:** Build instant messaging, live broadcast feeds, and push notification infrastructure on PHP/Laravel, NodeJS, WebSockets, and Redis backplanes.

Selected client engagements

CODOT — Distributed Digital Signage Platform · Full Stack Engineer & Cloud Architect [2024 – 2026]

- **Multi-region disaster recovery:** Led emergency migration from AWS Bahrain to AWS Mumbai after regional data centres were destroyed during the Iran conflict — rebuilt the full stack from scratch with no recoverable backups, restoring service and reconnecting 250+ devices across the fleet.
- **Sovereign storage workaround (Oman):** Designed and deployed custom MinIO object storage to serve the Oman region, where native AWS bucket support was unavailable, keeping media delivery uninterrupted across endpoints.
- **P2P streaming architecture:** Built a Node.js WebSocket server paired with a custom Coturn (TURN/STUN) deployment for low-latency P2P live streaming and bidirectional telemetry between the management dashboard and 250+ distributed player devices across Bahrain.
- **Native playback client:** Delivered the hybrid signage player, then rewrote it natively in Kotlin to gain hardware acceleration, reducing CPU usage on the player hardware.
- **Control plane:** Designed the backend APIs and administrative portal managing device state, ad workflows, and real-time playback monitoring at fleet scale.

Jamaheer (Bahrain Premier League) — Football Analytics Platform · Backend Development Lead [2026]

- **Infrastructure & launch:** Provisioned resilient VPS infrastructure and domain routing for a high-traffic live-score and analytics platform for the Bahraini market.
- **Real-time match architecture:** Designed backend systems for live match tracking and tournament data, integrating with cross-functional mobile teams; delivered squad lineup builder, live match control, and statistics generation using AI-assisted development to compress the build cycle to 8 weeks.

RCSI AI Simulator — Clinical Training Platform · Lead Full Stack & AI Integration Engineer [2024 - 2026]

- **Production LLM integration:** Worked alongside AI specialists to embed LLMs into a live clinical simulation environment driving real-time simulated patient dialogue and automated triage scoring for medical practitioners.
- **State-intensive backend:** Engineered the core Laravel infrastructure and tuned SQL schemas to sustain long-running, state-heavy simulation sessions for multiple concurrent candidates.
- **Frontend leadership:** Guided the React/Inertia.js SPA development with code review and mentorship; configured a Laravel Nova portal giving non-technical administrators self-serve control of candidates, scenarios, and assessment analytics under zero-downtime deploys.

AL MSAYYAN (Eskan Bank) — Land & Plot Management Platform · Senior Software Engineer [2024]

- **Interactive geospatial system:** Built a live land visualisation and plot management platform using Leaflet.js with custom map overlays and dynamic interactive polygons across 250+ residential plots.
- **Real-time state sync:** Implemented live polling to synchronise plot status and colour-coded availability, with Laravel Turbolinks for partial page loads; extended the admin dashboard so non-technical staff manage inventory and map coordinates without developer involvement.

Additional engagements · Vigilance Consulting · King Hamad Global Center (Bahrain) · Raed, Injaz Academy, LAX [2021 – 2026]

- Led Laravel/Statamic CMS architecture for two client platforms — migrating one legacy Vue/Laravel site with zero data loss, and building the second as a self-serve, no-code content environment from the ground up; owned production deployment via Laravel Forge.
- Delivered cross-platform EdTech and engagement mobile apps (Ionic, Angular, Laravel) for Middle Eastern markets, including enrollment, assessment, and scheduled push-notification pipelines.

Freelance Software Architect (concurrent) 2020 – Present

LetsUpDoc — Telemedicine Platform · End-to-End Architect [2020 – Present]

- **Full platform ownership:** Directed the complete lifecycle — patient portals, real-time video consultation, diagnostic lab results, pharmacy inquiry — across web and mobile, published to the App Store and Google Play.
- **Patient-facing LLM systems:** Built AIDO, a conversational health assistant, and a symptom checker that recommends specialists from reported symptoms — matching against physician specialty, bio, and procedure data embedded in Pinecone for natural-language search.
- **Longitudinal consultation memory:** Designed persistent storage and retrieval of patient–doctor consultation history, surfacing prior context to clinicians at follow-up so continuity of care carries across visits.
- **Infrastructure:** Ubuntu VPS provisioning with Nginx/SSL and zero-downtime GitHub Actions pipelines on Laravel Octane.

LetsUpMart — E-Commerce Platform · Platform Engineer [2025 – Present]

- **Catalog automation at scale:** Built Python scrapers and ETL pipelines ingesting 1000+ products from external sources, with an LLM post-processing stage that rewrites descriptions for uniqueness and SEO — replacing 10+ hours/week of manual cataloging.
- **Custom commerce architecture:** Heavily extended a Bagisto/Laravel base, re-engineering native admin workflows and building new operational pipelines for bespoke order management; led technical SEO and performance work. LLM-based catalog de-duplication resolved duplicate-content exclusions, taking indexed pages from 0 to 1000+.

EDUCATION

BSc Computer Science — NED University, Karachi · CGPA 3.51 2017 – 2020